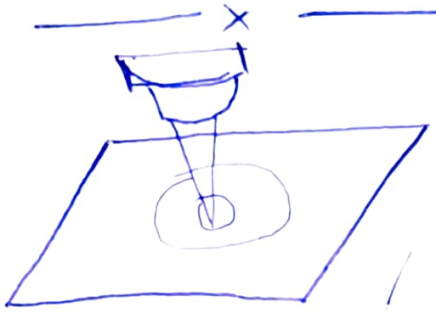
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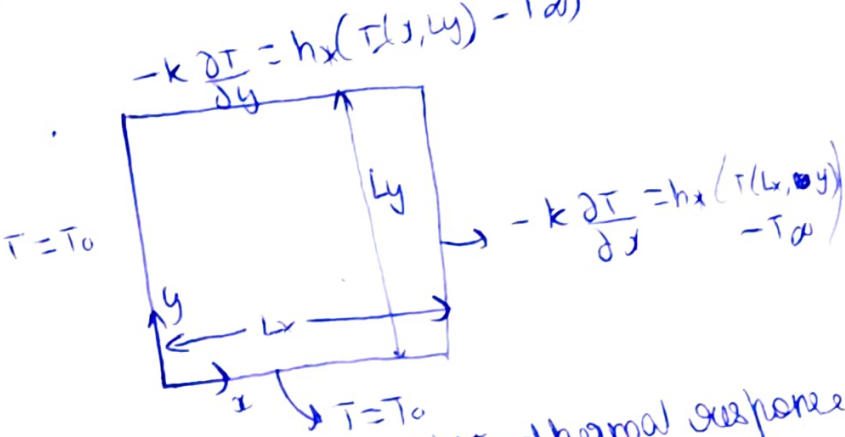
Computational Fluid Dynamics

Project -2

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(a) Problem Statement





→ In this project, our objective is to analyse steady state thermal response of a thin metallic plate being heated locally by a laser source.

→ Since the plate is thin, we can neglect temperature variation through its thickness and treat it as a 2D problem with internal heat generation

→ Boundary conditions: left and Bottom Edges are anchored at a constant baseline temperature (T_0). Top and Right boundaries are via convection. ($T(x,0) = T_0$, $T(0,y) = T_0$, $-k\frac{\partial T}{\partial y} = h_x(T(x,L_y) - T_\infty)$, $-k\frac{\partial T}{\partial x} = h_x(T(L_x,y) - T_\infty)$)

→ The governing balance equation combines 2D heat conduction with volumetric heat source (internal produced by laser)

$$q'''(x) = S_0 e^{\left\{-\frac{(x-x_0)^2 + (y-y_0)^2}{\delta^2}\right\}}$$

x_0, y_0 → coordinates of point where laser is pointed at)

→ In order to get a generalised solution, we have to non-dimensionalize the equation and express the variables in terms of non-dimensional variables)

→ We have to also solve a more general equation of form $-\nabla^2\phi = f_1(x,y)\phi + f_2(x,y)$ $0 < x < L_x$, $0 < y < L_y$

subject to B.C: $\phi(x=0) = \phi(0,y) = \phi_0$ $\left(\frac{\partial\phi}{\partial y}\right)_{x,L_y} + h_2\phi(x,L_y) = 0$

$$\left(\frac{\partial \phi}{\partial t}\right)_{L,y} + h_1 \phi(L,y) = 0$$

→ We have to develop FDM code to solve the above equation subject to the given boundary conditions.

→ As a check, we have to use the code to solve the given heat transfer problem

Energy Equation derivation

from RTT, we know

$$\left.\frac{dN}{dt}\right|_{sys} = \frac{\partial}{\partial t}\int_{\forall} \rho \phi \, d\forall + \int_{CS} (\rho \phi)\, \vec{V} \cdot \hat{n} \, dS$$

$\phi = e = e_{in} + e_{kn} + e_p$

→ CPT

$$\left.\frac{dN}{dt}\right|_{sys} = \dot{Q}_{in} - \dot{W}_{out}$$

$\dot{W}_{out}$ → W. done by sys → Work interaction

$\dot{Q}_{in}$ → heat interaction

$$\dot{Q}_{in,net} = -\int_{CS} q'' \cdot \hat{n} \, dS + \int_{CV} q'''(x) \, d\forall$$

$-\int_{CS}$ → Conduction flux; $\int_{CV}$ → Internal heat generation

$$W_{out} = -\dot{F}\cdot\underline{V} = -\int (\hat{n} \cdot \underline{\underline{\tau}}) \cdot \vec{V} \, dS$$

Since our system is stationary $\vec{V} = 0$ $\therefore \int \rho \phi \, (\vec{V} \cdot \hat{n}) \, dA = 0$

and $W_{out} = -\int (\hat{n} \cdot \underline{\underline{\tau}}) \cdot \vec{V} \, dS = 0$

Assuming a steady state system $\frac{\partial}{\partial t}\int_{\forall} \rho \phi \, d\forall = 0$

$\therefore$ RHS $= 0$

$$LHS = -\int_{CS} q'' \cdot \hat{n} \, dS + \int_{CV} q'''(x) \, d\forall = 0$$

Using Gauss Divergence Theorem, $-\int q'' \cdot \hat{n} \, dS = -\int (\nabla \cdot q'') \, d\forall$

$\therefore \int (q'''(x) - (\nabla \cdot q'')) \, d\forall = 0 \Rightarrow q'''(x) = \nabla \cdot q''$

Fourier's law $\Rightarrow q'' = -k \nabla T$

Substituting this, $q'''(x) = \nabla \cdot (-k \nabla T)$

$\therefore q'''(x) = -k \nabla \cdot (\nabla T)$

$\therefore q'''(x) = -k \nabla^2 T$

$\therefore k \nabla^2 T + q'''(x) = 0$ → Energy Equation (1)

Boundary Conditions: $T(x,0) = T_0$, $T(0,y) = T_0$, $\left(k \frac{\partial T}{\partial y}\right)_{x,y} + h_x (T(x,y) - T_\infty) = 0$

$k \left(\frac{\partial T}{\partial x}\right)_{L_x,y} + h_x (T(L_x,y) - T_\infty) = 0$

where $q'''(x) = S_0 e^{-\left(\frac{(x-x_0)^2 + (y-y_0)^2}{\sigma^2}\right)}$

Non-Dimensionalisation

$\theta = \frac{T(x) - T_\infty}{T_0 - T_\infty}$ (should use $T(x,y)$), $\bar{x} = \frac{x}{L_x}$, $\bar{y} = \frac{y}{L_y}$

$k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + q'''(x) = 0$ (should use $q'''(x,y)$)

$T(x) = (T_0 - T_\infty)\theta + T_\infty$

$\therefore \frac{\partial^2 T}{\partial x^2} = (T_0 - T_\infty) \frac{\partial^2 \theta}{\partial x^2}$

$\frac{\partial^2 T}{\partial x^2} = (T_0 - T_\infty) \frac{\partial^2 \theta}{\partial (\bar{x} L_x)^2}$

$\therefore \frac{\partial^2 T}{\partial x^2} = \frac{(T_0 - T_\infty)}{L_x^2} \frac{\partial^2 \theta}{\partial \bar{x}^2}$

Similarly $\frac{\partial^2 T}{\partial y^2} = \frac{T_0 - T_\infty}{L_x^2} \frac{\partial^2 \theta}{\partial \bar{y}^2}$

Substituting back in equation,

$k \left(\frac{T_0 - T_\infty}{L_x^2}\right)\left(\frac{\partial^2 \theta}{\partial \bar{x}^2} + \frac{\partial^2 \theta}{\partial \bar{y}^2}\right) + q'''(x) = 0$

Dividing by $k (T_0 - T_\infty) / L_x^2$

$\frac{\partial^2 \theta}{\partial \bar{x}^2} + \frac{\partial^2 \theta}{\partial \bar{y}^2} + \frac{q'''(x) L_x^2}{k (T_0 - T_\infty)} = 0$

Let us define dimensionless heat source term to be $\bar{q}'''(\bar{x}) = \frac{L_x^2}{k(T_0 - T_\infty)} q'''(x)$ (should use $\bar{q}'''(\bar{x}, \bar{y})$)

Original heat generation term

$$q'''(x) = \hat{S}_0 e^{-\left[\frac{(x-x_0)^2 + (y-y_0)^2}{\delta^2}\right]}$$

Substituting $x = \bar{x} L_x$, $y = \bar{y} L_x$, $x_0 = \bar{x}_0 L_x$, $y_0 = \bar{y}_0 L_x$ and $\delta = \bar{\delta} L_x$

and $q'''(\bar{x}) = \frac{L_x^2}{k(T_0 - T_\infty)} q'''(x)$

$$\frac{k(T_0-T_\infty)}{L_x^2} q'''(\bar{x}) = \hat{S}_0 e^{-\left[\frac{(\bar{x}-\bar{x}_0)^2 + (\bar{y}-\bar{y}_0)^2}{\delta^2}\right]}$$

$$\therefore q'''(\bar{x}) = \frac{\hat{S}_0 L_x^2}{k(T_0-T_\infty)} e^{-\left[\frac{(\bar{x}-\bar{x}_0)^2 + (\bar{y}-\bar{y}_0)^2}{\delta^2}\right]}$$

Substitute $S_0 = \frac{\hat{S}_0 L_x^2}{k(T_0-T_\infty)}$, $e^{-\left[\frac{(\bar{x}-\bar{x}_0)^2 + (\bar{y}-\bar{y}_0)^2}{\delta^2}\right]}$

Final Dimensionless form: should be $q'''(\bar{x}, \bar{y})$ where $0 < x < L_x$

$$\frac{\partial^2 \theta}{\partial \bar{x}^2} + \frac{\partial^2 \theta}{\partial \bar{y}^2} + q'''(\bar{x}) = 0$$

$\therefore 0 < \bar{x} L_x < L_x$

$0 < \bar{x} < 1$

where $q'''(\bar{x}) = S_0 e^{-\left[\frac{(\bar{x}-\bar{x}_0)^2 + (\bar{y}-\bar{y}_0)^2}{\delta^2}\right]}$ where $0 < y < L_y$

$0 < \bar{y} L_x < L_y$

$\therefore 0 < \bar{y} < L_y / L_x = \bar{L}_y$

Boundary Conditions non-dimensionalisation

(1) $T(x,0) = T_0 \Rightarrow \theta(\bar{x}, 0) = \frac{T(x,0) - T_\infty}{T_0 - T_\infty} = \frac{T_0 - T_\infty}{T_0 - T_\infty} = 1$

(2) $T(0,y) = T_0 \Rightarrow \theta(0, \bar{y}) = \frac{T(0,y) - T_\infty}{T_0 - T_\infty} = \frac{T_0 - T_\infty}{T_0 - T_\infty} = 1$

(3) $\frac{\partial T}{\partial y} = \frac{(T_0 - T_\infty)}{L_x} \frac{\partial \theta}{\partial \bar{y}}$ (for top boundary, $\bar{y} = \bar{L}_y$)

$$\therefore -\frac{k(T_0 - T_\infty)}{L_x} \left(\frac{\partial \theta}{\partial \bar{y}}\right)_{\bar{x}, \bar{y} = \bar{L}_y} = h_* (T_0 - T_\infty) \theta(\bar{x}, \bar{L}_y)$$

$$\therefore \left(\frac{\partial \theta}{\partial \bar{y}}\right)_{\bar{x}, \bar{y} = \bar{L}_y} = -\frac{h_* L_x}{k} \theta(\bar{x}, \bar{L}_y)$$

$$\therefore \left(\frac{\partial \theta}{\partial \bar{y}}\right)_{\bar{x}, \bar{y} = \bar{L}_y} + \frac{h_* L_x}{k} \theta(\bar{x}, \bar{L}_y) = 0$$

$\Rightarrow h$

(4) $\frac{\partial T}{\partial x} = \frac{T_0 - T_\infty}{L_x} \frac{\partial \theta}{\partial \bar{x}}$ (for right boundary, $\bar{x} = 1$)

$-k \frac{(T_0 - T_\infty)}{L_x} \left(\frac{\partial \theta}{\partial \bar{x}}\right)_{1,\bar{y}} = h_x (T_0 - T_\infty)\, \theta(1, \bar{y}) = 0$

$\therefore \left(\frac{\partial \theta}{\partial \bar{x}}\right)_{1,\bar{y}} = -\frac{h_x L_x}{k} \theta(1, \bar{y})$

$\therefore \left(\frac{\partial \theta}{\partial \bar{x}}\right)_{1,\bar{y}} + \underbrace{\frac{h_x L_x}{k}}_{=h} \theta(1, \bar{y}) = 0$

We have to solve general equation of form:

$\nabla^2 \phi = f_1(x,y)\, \phi + f_2(x,y) \quad 0 \le x \le L_x, \; 0 \le y \le L_y$

B.C $\quad \phi(x,0) = \phi(0,y) = \phi_0$

$\left(\frac{\partial \phi}{\partial y}\right)_{x, L_y} + h_2 \phi(x, L_y) = 0$

$\left(\frac{\partial \phi}{\partial x}\right)_{L_x, y} + h_1 \phi(L_x, y) = 0$

Mesh details and approach for discretization

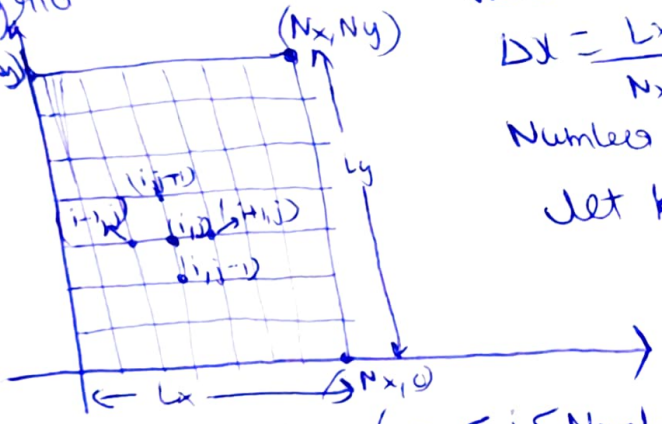
(b) Since its a 2D problem, our computational domain will be a 2D grid

We will choose a uniform grid in x and y

$\Delta x = \frac{L_x}{N_x - 1}, \quad \Delta y = \frac{L_y}{N_y - 1}$

Number of grid points $= N_x \times N_y$

Let $k = (j-1) N_x + i$



For unknown nodes, $(2 \le i \le N_x - 1, \; 2 \le j \le N_y - 1)$, the governing equation will be $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = f_1(x,y)\, \phi + f_2(x,y)$

for pt. (i,j) $\quad \phi \to \phi_{i,j}$

$x_i \to (i-1)\Delta x, \; y_j \to (j-1)\Delta y$

$f_1(x,y) = f_1(x_i, y_j)$

$f_2(x,y) = f_2(x_i, y_j)$

For points on bottom boundary, $j=1$, $i=1 \longrightarrow N_x$

governing equation: $\phi = \phi_0$

For points on left boundary, $i=1$, $j=1 \longrightarrow N_y$

governing equation, $\phi = \phi_0$

For points on top boundary, $j=N_y$, $i=1 \longrightarrow N_x$

governing equation: $\frac{\partial \phi}{\partial y} + h_2 \phi = 0$

For points on right boundary, $i=N_x$, $j=1 \longrightarrow N_y$

governing equation: $\frac{\partial \phi}{\partial x} + h_1 \phi = 0$

for the unknown nodes, for laplacian term, we will use second order CDS schemes for approximation.

For the points on boundaries whose governing equation has first derivative term, we will take care that while using first order approximation, we don't use points that is out of bound of the grid.

(c) Derivation and Presentation of final forms of discretized equations

for unknown points: $(2 \le i \le N_x, 2 \le j \le N_y)$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = f_1(x,y)\,\phi + f_2(x,y)$$

$k = (j-1)N_x + i$

$$\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{\Delta x^2} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta y^2} = f_1(x_i, y_j)\,\phi_{i,j} + f_2(x_i, y_j)$$

$$\frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta x^2}$$

multiply by Δy^2,

$$\Delta y^2 \left(\frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta x^2} \right) + \phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1} = f_1(x_i, y_j)\,\phi_{i,j}\,\Delta y^2 + f_2(x_i, y_j)\,\Delta y^2$$

$$A_E \phi_{i+1,j} + A_W \phi_{i-1,j} + A_P \phi_{i,j} + A_N \phi_{i,j+1} + A_S \phi_{i,j-1} = Q_{i,j}$$

$A_E = \Delta y^2/\Delta x^2$, $A_W = \Delta y^2/\Delta x^2$, $A_P = -\frac{2\Delta y^2}{\Delta x^2} - 2 - f_1(x_i, y_j)\Delta y^2$, $A_N = A_S = 1$,

$Q_{i,j} = f_2(x_i, y_j)\Delta y^2$

for bottom boundary ($j=1$, $i=1 \to Nx$), $k=i$

$A(k,k) = Ap$, $A(k,k+1) = AE$, $A(k,k+Nx) = AN$, $A(k,k-1) = Aw$, $A(k,k-Nx) = As$

for bottom boundary ($j=1$, $i=1 \to Nx$)

$k = i$

$\phi_k = \phi_0$

for left boundary ($i=1$, $j=2 \to Ny$)

$k = (j-1)Nx + 1$

$\phi_k = \phi_0$

for top boundary, ($j=Ny$, $i=2 \to Nx$)

$k = (Ny-1)Nx + i$

$$\frac{\partial \phi}{\partial y} + h_2 \phi = 0$$

$$\therefore \frac{\phi_{i,j} - \phi_{i,j-1}}{\Delta y} + h_2 \phi_{i,j} = 0$$

$$\therefore (1 + h_2 \Delta y)\, \phi_{i,j} - \phi_{i,j-1} = 0$$

$j = Ny$

$$\therefore (1 + h_2 \Delta y)\, \phi_{i,Ny} - \phi_{i,Ny-1} = 0$$

$$Ap\, \phi_{i,Ny} + As\, \phi_{i,Ny-1} = 0$$

$Ap = (1 + h_2 \Delta y)$, $As = -1$

for right boundary, ($i=Nx$, $j=2 \to Ny-1$)

$$\frac{\partial \phi}{\partial x} + h_1 \phi = 0, \quad k = (j-1)Nx + Nx = jNx$$

$$\frac{\phi_{i,j} - \phi_{i-1,j}}{\Delta x} + h_1 \phi_{i,j} = 0$$

$$\phi_{i,j} - \phi_{i-1,j} + h_1 \Delta x\, \phi_{i,j} = 0$$

$$(1 + h_1 \Delta x)\, \phi_{i,j} - \phi_{i-1,j} = 0$$

$$(1 + h_1 \Delta x)\, \phi_{Nx,j} - \phi_{Nx-1,j} = 0$$

$$Ap\, \phi_{Nx,j} + Aw\, \phi_{Nx-1,j} = 0$$

$Ap = 1 + h_1 \Delta x$, $Aw = -1$

(d) Solution Methodology

As a test case, we have to solve the given heat transfer problem subject to the given boundary conditions.
We have already derived discretized form of equations for the general form.
So first we will code the fdm solver for the general equation.
This will be a separate file.
It will contain a function general_FDM_solver having inputs → $L_x, L_y, N_x, N_y, f_1, f_2$ (function handles), h_1, h_2, ϕ_0.

Then we will form the coefficient matrix A by looping and using the governing equations for the respective points. Similarly we will form the Q matrix (rhs_val in code).
Then we will compute $A^{-1}Q$ to find our ϕ matrix and reshape it into an matrix $\phi(i,j)$.
outputs → $\phi(i,j)$, x and y grid vectors.

File 2 → test case solver
→ We will pick appropriate choice of $L_x, L_y, N_x, N_y, h, \phi_0, S_0, x_0, y_0$ to solve for the heat problem
→ $f_1 = 0$ and $f_2 = -q'''(\bar{x}, \bar{y}) = -S_0 e^{-\left[\frac{(\bar{x}-\bar{x}_0)^2 + (\bar{y}-\bar{y}_0)^2}{\delta^2}\right]}$
→ will call the general_FDM_solver
⇒ obtain $\phi(\bar{x}, \bar{y})$ and $\bar{x}, \bar{y}$ grid mat.
→ Plot $\phi(\bar{x}, \bar{y})$ contours, line plots ϕ v/s x and ϕ v/s y

$0 < \bar{x} < 1$
$0 < \bar{y} < \frac{L_y}{L_x}$

Results and Discussions

The FDM solver was applied with $L_x = 1$, $L_y = 1.5$ and $h_1 = h_2$ on the top and right edges, $S_0 = 100$, $x_0 \approx 0.5$, $y_0 \approx 0.5$ and $\delta = 0.1$

→ Overall field → The contour plot shows a localised hot region peaking near the source location, decaying outward & cools the rectangular domain. There is elongation in y as the hot core occupies a smaller fraction of domain's total height and cooling towards the top edge is significantly larger than towards the right region.

→ As $h_1 = h_2$, the right and top edges cool the plate at different rates which is evident by looking in the line plots. along the top edge, the curves in the ϕ v/s y plot fall to noticeably lower values (down to 0.2 as to 0.4) than values reached at the right edge in the ϕ v/s x plot for the similar region (0.4 to 0.1). This indicates that heat is lost more aggressively through top boundary than right boundary.

→ Curves passing close to source's coordinates show a sharp and tall peak while curves corresponding to further regions show a flatter decreasing profile.

→ The peak in ϕ v/s y plot occurs at $y = 0.5$, which is < 1.5
∴ A substantial fraction of domain lies outside the source's direct influence

Conclusion: 1) We developed FDM solver for general PDE:
$\nabla^2 \phi = f_1(x,y)\,\phi + f_2(x,y)$ subject to Dirichlet and Robin boundary conditions

2) The solver was applied to the specific laser heating problem, and contours, ϕ v/s x, ϕ v/s y was plotted and was visually validated by observing contour plot of B.C matched the contours plot as well it was seen that maximum temperature was obtained near the laser source